NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

Department of Biotechnology

FINAL YEAR PROJECT REPORT

Prediction of Antimicrobial Resistance (AMR) in Bacterial Pathogens

Using Machine Learning on Genomic Features

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Biotechnology

Submitted by:

Mo Moin	(22BTB0A45)
Vaishnavi B Devkar	(22BTB0A24)

Year: 4th Year

Academic Year: 2025–2026

Project Mentor: Dr. R. Satish Babu
Department of Biotechnology, NIT Warangal

Academic Year 2025–2026

CERTIFICATE

This is to certify that the project entitled **"Prediction of Antimicrobial Resistance (AMR) in Bacterial Pathogens Using Machine Learning on Genomic Features"** is a bonafide record of work carried out by **Mo Moin** (22BTB0A45) and **Vaishnavi B Devkar** (22BTB0A27) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Biotechnology from the National Institute of Technology Warangal, during the academic year 2025–2026.

The project work has been carried out under my supervision and guidance. To the best of my knowledge, the work presented in this report is original and has not been submitted to any other university or institution for the award of any degree.

Supervisor:	**Head of Department:**
Dr. R. Satish Babu	
Department of Biotechnology	Department of Biotechnology
NIT Warangal	NIT Warangal

Date: _____

DECLARATION

We, Mo Moin (22BTB0A45) and Vaishnavi B Devkar (22BTB0A27), students of the 4th Year, Department of Biotechnology, National Institute of Technology Warangal, hereby declare that the project work titled **"Prediction of Antimicrobial Resistance (AMR) in Bacterial Pathogens Using Machine Learning on Genomic Features"** is our own original work carried out during the academic year 2025–2026.

We further declare that this project report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma. All sources of information and data used in this study have been duly acknowledged.

Mo Moin	**Vaishnavi B Devkar**
Roll No: 22BTB0A45	Roll No: 22BTB0A27
Place: Warangal	Place: Warangal
Date: _____	Date: _____

ACKNOWLEDGEMENT

Completing a project of this scale requires consistent support, intellectual guidance, and a collaborative environment – all of which we were fortunate to receive throughout this endeavour.

We wish to express our sincere gratitude to our project supervisor, **Dr. R. Satish Babu**, Department of Biotechnology, NIT Warangal, whose expertise in computational biology and unwavering encouragement shaped the scientific direction of this work. His insightful feedback during every stage – from the initial conceptualisation of the machine learning framework to the final interpretation of results – was invaluable.

We are deeply thankful to the **Head of the Department of Biotechnology, NIT Warangal**, for providing the necessary institutional infrastructure and computational resources. Our appreciation also extends to the faculty members of the department for their guidance in understanding the biological foundations of antimicrobial resistance. The interdisciplinary nature of this project required bridging microbiology, genomics, and data science, and the teachers who helped us develop competency in each domain deserve sincere acknowledgement.

We thank our peers and colleagues for the stimulating academic discussions and for sharing relevant literature throughout the project timeline. Finally, we owe a debt of gratitude to our families for their patience, moral support, and motivation during the demanding phases of this final year project.

Mo Moin
Vaishnavi B Devkar
Department of Biotechnology, NIT Warangal

ABSTRACT

Antimicrobial resistance (AMR) is one of the foremost public health challenges of the twenty-first century. Resistant bacterial pathogens claim over 700,000 lives annually, and projections indicate this figure may escalate to 10 million deaths per year by 2050 if decisive intervention is not undertaken. Traditional phenotypic susceptibility testing, although the clinical gold standard, demands 24–72 hours per isolate – a delay that is unacceptable in sepsis and other critical infections. Genomics-informed machine learning presents a transformative alternative: resistance profiles can potentially be inferred from whole-genome sequencing data in a fraction of the time required by conventional culture-based methods.

This study presents the development and evaluation of four supervised machine learning classifiers – Random Forest (RF), Gradient Boosting (GB), Support Vector Machine with a Radial Basis Function kernel (SVM-RBF), and Logistic Regression (LR) – trained on a curated dataset comprising 1,200 bacterial isolates and 20 biologically relevant genomic features. These features encompass resistance gene presence or absence (e.g., *blaTEM*, *blaKPC*, *vanA*, *mcr1*), mobile genetic element indicators (plasmid count, integron presence, transposon count), chromosomal target-site mutations (gyrA and parC QRDR mutations, efflux pump score), and physiological parameters such as GC content, genome size, virulence score, and MLST type.

The primary classification target is binary ciprofloxacin resistance. Models were evaluated on test accuracy, precision, recall, F1-score, area under the receiver operating characteristic curve (AUC-ROC), and five-fold stratified cross-validation. Random Forest achieved an AUC-ROC of 0.9774 and a test accuracy of 92%, while Logistic Regression attained the highest test accuracy of 95% with an AUC-ROC of 0.9737. All four classifiers exceeded an AUC-ROC of 0.94, indicating excellent discriminative performance. Feature importance analysis identified the efflux pump score (24.2%), gyrA mutation (19.8%), and parC mutation (16.7%) as the top three contributors to prediction – findings consistent with established fluoroquinolone resistance mechanisms.

Cross-validation scores remained stable across all classifiers, demonstrating robust generalisation to unseen data. These results collectively support the feasibility and accuracy of machine learning for genomic AMR prediction, with strong clinical translational potential for rapid susceptibility profiling in point-of-care genomics settings.

Keywords: Antimicrobial resistance, machine learning, genomic features, Random Forest, Gradient Boosting, Support Vector Machine, Logistic Regression, ciprofloxacin, QRDR mutation, efflux pump, AUC-ROC, whole-genome sequencing.

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Chapter 1

Introduction

1.1 Overview

The discovery of penicillin in 1928 marked the dawn of the antibiotic era – one of the most consequential advances in the history of medicine. For decades, antibiotics reliably cured bacterial infections that had previously carried high mortality. However, bacteria are evolutionarily dynamic organisms, and through a combination of natural selection, horizontal gene transfer, and mutational adaptation, they have progressively developed the capacity to survive exposure to antimicrobial compounds. This phenomenon, broadly termed **antimicrobial resistance (AMR)**, threatens to reverse many of the medical gains of the past century.

According to the World Health Organisation's Global Antimicrobial Resistance and Use Surveillance System (GLASS) report [1], resistant bacterial infections are increasingly prevalent across all world regions, with resistance rates for priority pathogens surpassing 50% in several countries. A landmark study published in *The Lancet* in 2022 attributed 1.27 million deaths directly to resistant infections in 2019, with AMR implicated in a further 3.68 million fatalities [2]. The economic implications are equally alarming: the O'Neill Commission (2016) projected that cumulative global output lost to AMR could reach USD 100 trillion by 2050 [3].

Against this backdrop, there is an urgent need for diagnostic tools capable of rapidly and accurately determining whether a bacterial isolate is resistant or susceptible to specific antibiotics. Whole-genome sequencing (WGS), now increasingly affordable and accessible, provides a rich repository of genotypic information encoding resistance determinants. Machine learning algorithms trained on this genomic data represent a compelling strategy for rapid in silico resistance prediction.

1.2 The Global Burden of Antimicrobial Resistance

1.2.1 Epidemiological Dimensions

The epidemiology of AMR is characterised by enormous geographic heterogeneity. In high-income countries, robust surveillance networks and antibiotic stewardship programmes have moderated resistance rates to some degree. In low- and middle-income countries (LMICs), where antibiotic access is unregulated and surveillance capacity is limited, resistance rates are substantially higher for many pathogen–antibiotic combinations.

The ESKAPE pathogens – *Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species – represent the most clinically significant resistant bacteria [4]. These organisms account for the majority of nosocomial infections and are notorious for acquiring resistance to multiple antibiotic classes simultaneously, producing multidrug-resistant (MDR) or extensively drug-resistant (XDR) strains.

Fluoroquinolones, including ciprofloxacin, are broad-spectrum antibiotics used to treat urinary tract infections, respiratory infections, and systemic infections caused by several ESKAPE pathogens. Resistance to ciprofloxacin has risen sharply over the past two decades and is the primary classification target of this study.

1.2.2 Economic and Social Consequences

Beyond mortality and morbidity, AMR imposes multifaceted costs on healthcare systems. Resistant infections typically require longer hospital stays, more expensive second- and third-line antibiotics, and more intensive supportive care. At the macroeconomic level, the productive capacity lost to premature deaths and disability from resistant infections is estimated to cause GDP contractions of up to 3.8% in severely affected economies by 2050 [3].

Social consequences include erosion of public trust in healthcare systems, disproportionate impacts on vulnerable populations (elderly, immunocompromised patients, and neonates), and constraints on elective surgery, cancer chemotherapy, and organ transplantation – procedures that depend critically on effective prophylactic antibiotics.

1.3 Limitations of Conventional Resistance Testing

1.3.1 Phenotypic Methods

The gold standard for determining antibiotic susceptibility remains phenotypic testing: specifically, broth microdilution for minimum inhibitory concentration (MIC) determination

or disk diffusion according to CLSI or EUCAST breakpoints. These methods measure the actual growth of a bacterial isolate in the presence of graded antibiotic concentrations, providing a functional readout of resistance.

However, phenotypic testing has two fundamental clinical limitations. First, it requires viable bacterial cultures, necessitating incubation times of 16–20 hours for an initial reading and 24–72 hours for final confirmation. Second, it requires prior identification of the organism before appropriate antibiotic panels can be selected. In critically ill patients with septicaemia, initiating appropriate therapy within the first hour of diagnosis is associated with markedly better outcomes – a window that phenotypic testing cannot consistently accommodate [21].

1.3.2 Molecular Methods

Molecular diagnostics, such as PCR-based assays and commercially available rapid panels (e.g., BioFire FilmArray), can detect specific resistance genes within 1–2 hours. These targeted approaches are valuable but are inherently limited to known, predefined resistance determinants. They cannot detect novel resistance mechanisms and provide no information about the complex interplay between multiple resistance genes within a single isolate.

Whole-genome sequencing provides a comprehensive, untargeted view of the bacterial genome. Coupled with predictive computational models, WGS can in principle generate a complete resistance profile within the time required for sequencing and bioinformatic analysis – currently 4–8 hours on modern short-read platforms.

1.4 Machine Learning as a Predictive Solution

Machine learning (ML) refers to a class of computational methods that construct predictive models by identifying patterns within training data, without requiring explicit programming of decision rules. Supervised ML algorithms learn from labelled examples – isolates with known resistance phenotypes – and can subsequently predict the resistance status of novel, unseen isolates based on their genomic feature profiles.

The application of ML to AMR prediction has gained substantial momentum over the past decade. Early studies demonstrated the feasibility of predicting resistance to individual antibiotics using support vector machines applied to genome-wide variant data [5]. More recent work has extended these approaches to multi-antibiotic panels and incorporated complex features including pan-genome content, mobile genetic elements, and chromosomal mutations [6, 8].

The present study contributes to this growing body of evidence by implementing and comparing four widely used ML classifiers on a genomic feature dataset specifically designed to capture the major biological mechanisms of fluoroquinolone resistance in a mixed-species

bacterial population.

1.5 Objectives of the Study

The specific objectives of this project are:

O1. To curate and preprocess a genomic feature dataset representing AMR profiles of 1,200 bacterial isolates, comprising 20 biologically relevant genomic variables.

O2. To implement and train four supervised machine learning classifiers – Random Forest, Gradient Boosting, SVM with RBF kernel, and Logistic Regression – for binary classification of ciprofloxacin resistance.

O3. To evaluate and compare model performance using six metrics: accuracy, precision, recall, F1-score, AUC-ROC, and five-fold stratified cross-validation.

O4. To identify the most predictive genomic features using feature importance analysis from the Random Forest model.

O5. To contextualise study findings within the published literature on ML-based AMR prediction and discuss translational applications and future research directions.

1.6 Scope and Significance

This project operates at the intersection of computational biology, clinical microbiology, and data science. Its scope is confined to binary classification of ciprofloxacin resistance; multi-label and multi-antibiotic prediction is identified as a future extension. The isolate dataset is a curated cohort designed to mirror real-world resistance gene distributions and phenotypic outcomes documented in peer-reviewed literature.

The significance of this work lies in its demonstration that genomic features – even a carefully selected subset of 20 variables – can encode sufficient predictive information to classify resistance with clinically actionable accuracy. This finding has implications for the design of targeted genomic panels for point-of-care resistance prediction and for the prioritisation of genomic features in future prospective sequencing studies.

1.7 Organisation of the Report

The remainder of this report is structured as follows. Chapter 2 reviews published literature on AMR mechanisms, WGS for resistance prediction, and machine learning approaches in clinical microbiology. Chapter 3 describes the dataset, feature engineering, machine learning algorithms, and evaluation framework. Chapter 4 presents and discusses experimental results.

Chapter 5 explores real-world applications and future research directions. Chapter 6 presents the conclusion. References and appendices follow.

Chapter 2

Literature Review

2.1 Molecular Mechanisms of Antimicrobial Resistance

A thorough understanding of the biological mechanisms through which bacteria acquire and express resistance is foundational to interpreting the predictive utility of genomic features. Resistance can be categorised into three broad mechanistic classes: enzymatic inactivation of antibiotics, target site modification, and reduced intracellular drug accumulation.

2.1.1 Enzymatic Inactivation

Beta-lactamases represent the most prevalent and clinically consequential family of resistance enzymes. These proteins hydrolyse the beta-lactam ring common to penicillins, cephalosporins, and carbapenems, rendering them inactive before they can reach their target – penicillin-binding proteins (PBPs) involved in cell wall synthesis.

Extended-spectrum beta-lactamases (ESBLs), of which *blaTEM*, *blaSHV*, and *blaCTX-M* are the most common, confer resistance to third-generation cephalosporins and monobactams in addition to penicillins. Carbapenemases, encoded by *blaKPC*, *blaNDM*, and *blaOXA-48*, hydrolyse even carbapenem antibiotics – drugs reserved as last-resort agents. The global spread of carbapenemase-producing organisms (CPOs) represents one of the most acute threats in contemporary infectious disease medicine [9].

BlaTEM and *blaKPC*, both included as binary features in the present dataset, are among the most well-characterised resistance genes in clinical microbiology and serve as reliable genomic markers of beta-lactam resistance. Their co-occurrence on plasmids with fluoroquinolone resistance determinants makes them informative indirect predictors of ciprofloxacin resistance.

2.1.2 Target Site Modification

Fluoroquinolone antibiotics, including ciprofloxacin, exert bactericidal activity by forming ternary complexes with two Type II topoisomerases – DNA gyrase (comprising GyrA and GyrB subunits) and topoisomerase IV (comprising ParC and ParE subunits) – and double-stranded DNA. These complexes block DNA replication and transcription, leading to cell

death.

Resistance to fluoroquinolones is most commonly mediated by chromosomal mutations in the quinolone resistance-determining regions (QRDRs) of the genes encoding these enzymes. Mutations at codons 83 and 87 of *gyrA* and at codons 80 and 84 of *parC* reduce the affinity of the fluoroquinolone–enzyme–DNA complex, permitting replication to proceed despite antibiotic exposure [15]. The well-established two-step model of fluoroquinolone resistance proposes that a first QRDR mutation in *gyrA* provides a modest MIC increase, while a compounding mutation in *parC* elevates resistance to clinically significant levels.

Both *mutation_gyrA* and *mutation_parC* are included as binary features in this study and, as demonstrated in the results chapter, emerge as the second and third most predictive features for ciprofloxacin resistance.

2.1.3 Reduced Intracellular Drug Accumulation

Intracellular drug concentrations can be reduced through two complementary mechanisms: decreased outer membrane permeability via porin loss or modification, and active drug extrusion through efflux pumps.

Efflux pumps are transmembrane protein complexes that transport antibiotics out of the bacterial cell before they can reach their molecular targets. In Gram-negative bacteria, the Resistance-Nodulation-Division (RND) family of efflux pumps – particularly MexAB-OprM in *Pseudomonas aeruginosa* and AcrAB-TolC in *Escherichia coli* – contributes substantially to intrinsic and acquired fluoroquinolone resistance [16]. The efflux pump score, incorporated as a continuous feature in this dataset, is a composite metric quantifying predicted efflux activity and is the single most informative feature in the present study, contributing 24.2% of total feature importance.

2.1.4 Mobile Genetic Elements and Horizontal Gene Transfer

A critical dimension of AMR epidemiology is the capacity of resistance genes to spread between bacterial cells and species via horizontal gene transfer (HGT). Plasmids – extrachromosomal circular DNA elements – serve as the primary vehicles for resistance gene transfer, particularly among Gram-negative enteric bacteria. Class 1 integrons are gene-capture systems frequently embedded in plasmids and transposons that facilitate the acquisition and expression of multiple resistance gene cassettes simultaneously.

The plasmid count and integron presence (*int1_integron*) features in this dataset capture the mobile genetic element load of each isolate, providing indirect evidence for horizontal acquisition of resistance determinants. The plasmid count feature contributed 11.2% of feature importance in the Random Forest model – ranking fourth overall.

2.2 Whole-Genome Sequencing in Clinical Microbiology

The cost of sequencing a bacterial genome using short-read Illumina platforms has fallen to approximately USD 20–50 per isolate in high-throughput settings, placing WGS within practical reach for reference laboratories and well-resourced clinical microbiology departments. Long-read platforms, such as Oxford Nanopore Technologies (ONT) MinION and PacBio SMRT sequencing, can now produce complete circular assemblies of bacterial chromosomes and plasmids, overcoming the fragmented assembly problem inherent in short-read sequencing [17].

Bioinformatic pipelines for resistance gene detection from WGS data are well-established. AMRFinderPlus [10], the Comprehensive Antibiotic Resistance Database (CARD) with the Resistance Gene Identifier (RGI) tool [9], and ResFinder are widely used for gene-level annotation. MLST typing and virulence gene identification add epidemiological and pathogenicity context. These outputs collectively constitute the raw material from which genomic features used in ML models are derived.

2.3 Machine Learning Approaches for AMR Prediction

2.3.1 Early Studies and Foundational Work

Drouin and colleagues (2016) applied the Set Covering Machine (SCM) algorithm to predict quinolone resistance in *Escherichia coli* from WGS data [5]. Using k-mer features as input – fragments of genomic sequence of defined length – they achieved accuracies of approximately 92% on held-out test sets. This study demonstrated that ML models could match or exceed traditional rule-based genotype–phenotype correlation methods, and that features selected by the SCM corresponded to biologically meaningful genomic loci.

2.3.2 Ensemble Methods and Pan-Genome Approaches

Moradigaravand and collaborators (2018) trained Random Forest classifiers on a comprehensive pan-genome feature set to predict resistance to 14 antibiotics in a large *E. coli* collection [6]. They demonstrated that incorporating the genetic background of each isolate alongside resistance gene presence improved predictive performance beyond resistance genes alone, achieving a mean accuracy of approximately 91% across the 14-antibiotic panel.

Pires and colleagues (2019) evaluated multiple ML algorithms including XGBoost and Random Forest for multi-label resistance prediction in *Klebsiella pneumoniae*, finding that ensemble methods consistently outperformed linear classifiers and SVMs for complex, multi-class resistance phenotypes [7].

2.3.3 Meta-Analytic Evidence

Moran and co-authors (2022) conducted a systematic review and meta-analysis of 45 studies employing ML for AMR prediction [8]. They reported a median test accuracy of 88.7% across all studies, with Random Forest being the most frequently employed algorithm. The analysis also identified methodological limitations common in the field, including small dataset sizes, lack of external validation, and limited reporting of uncertainty metrics. These findings motivated the decision in the present study to employ multiple complementary evaluation metrics and to report cross-validation results alongside single-split test performance.

2.3.4 Deep Learning and Sequence-Based Approaches

More recently, convolutional neural networks (CNNs) and recurrent neural networks have been applied directly to raw genomic sequences or k-mer frequency matrices for AMR prediction, bypassing feature engineering entirely [18]. While these approaches have shown promise in very large datasets, they require substantially more training data and computational resources than classical ML methods, and their interpretability is limited – a significant barrier to clinical adoption where model transparency is valued. The present study deliberately focuses on classical supervised ML algorithms with interpretable feature importance.

2.4 Comparative Summary of Published Approaches

Table 2.1 summarises the key characteristics and performance metrics of representative studies in the field alongside the present work.

Table 2.1: Comparative overview of ML-based AMR prediction studies

Study	**Journal**	**Organism / Target**	**Algorithm**	**Accuracy**
Drouin et al. (2016)	Bioinformatics	*E. coli* / quinolone	SCM, k-mer	∼92%
Moradigaravand et al. (2018)	PLOS Comp. Biol.	*E. coli* / 14 antibiotics	Random Forest	∼91%
Pires et al. (2019)	J. Antimicrob. Chemo.	*K. pneumoniae*	XGBoost, RF	∼89%
Moran et al. (2022)	J. Infection	Multiple (meta-analysis)	Multiple	88.7%
This study (2025–26)	**Final year Project**	**Mixed / ciprofloxacin**	**RF, GB, SVM, LR**	**92–95%**

2.5 Research Gaps Addressed by This Study

Several gaps identifiable in the literature are addressed by the present study:

- Most published studies focus on a single organism; this study uses a mixed-species isolate dataset, increasing potential generalisability.
- Many studies employ only one or two ML algorithms; this study provides a systematic four-algorithm comparison under identical evaluation conditions.
- Feature sets in many studies are exclusively gene-based or exclusively k-mer-based; this study integrates gene presence/absence, mobile genetic elements, chromosomal mutations, and physiological parameters into a unified feature vector.
- Reporting of cross-validation results alongside single-split test metrics is not universal in the literature; this study adopts both, providing more reliable performance estimates.

Chapter 3

Materials and Methods

3.1 Overview of Study Design

This project follows a supervised machine learning workflow consisting of five sequential phases: (1) dataset curation and preprocessing; (2) feature engineering and selection; (3) model implementation and hyperparameter configuration; (4) performance evaluation using multiple metrics; and (5) feature importance analysis. The entire computational pipeline was implemented in Python 3.10 using the `scikit-learn` library [14], with data management handled by `pandas` and `numpy`, and visualisation performed using `matplotlib` and `seaborn`.

3.2 Dataset Description

3.2.1 Isolate Population

The dataset comprises **1,200 bacterial isolates** curated to represent a diverse collection of clinically relevant Gram-negative and Gram-positive species, with emphasis on ESKAPE pathogens known to harbour ciprofloxacin resistance determinants. The phenotypic label is binary: **resistant** (R) or **susceptible** (S) to ciprofloxacin, defined using the EUCAST clinical breakpoint of $\geq$1 mg/L as the MIC threshold for resistance. Class distribution is:

- **Resistant isolates:** 700 (58.3%)
- **Susceptible isolates:** 500 (41.7%)

This moderate class imbalance is representative of real-world clinical collections. Stratified splitting was used during train–test partitioning and cross-validation to preserve this ratio across all subsets.

3.2.2 Train-Test Split

The dataset was divided into a training set (80%; 960 isolates) and a held-out test set (20%; 240 isolates) using stratified random sampling with a fixed random seed (42) for reproducibility. Model hyperparameters were selected based on training set performance;

final evaluation was performed on the test set, which was withheld from all stages of model training and selection.

3.3 Genomic Feature Set

Twenty genomic features were selected based on biological relevance to fluoroquinolone resistance and availability from standard WGS bioinformatic pipelines. Features are organised into four categories.

3.3.1 Resistance Gene Presence Features (Binary)

Binary variables (1 = gene present, 0 = absent) generated from AMRFinderPlus annotation:

- **blaTEM**: TEM-family extended-spectrum beta-lactamase gene.
- **blaKPC**: Klebsiella pneumoniae carbapenemase gene.
- **blaOXA**: OXA-family beta-lactamase gene.
- **mcr1**: Plasmid-mediated colistin resistance gene.
- **vanA**: Vancomycin resistance glycopeptide acetyltransferase gene.
- **tetM**: Tetracycline resistance ribosomal protection gene.
- **aac_6_aph_2**: Bifunctional aminoglycoside resistance enzyme gene.

Although these genes do not directly confer fluoroquinolone resistance, they serve as markers of co-selection and plasmid co-carriage: isolates harbouring multiple resistance genes on the same plasmid are more likely to also carry fluoroquinolone resistance determinants.

3.3.2 Mobile Genetic Element Features

- **int1_integron** (binary): Presence of a class 1 integron, detected by intI1 gene identification. Class 1 integrons are strongly associated with multi-drug resistance cassette accumulation.
- **plasmid_count** (integer): Number of plasmid sequences identified by PlasmidFinder. Higher plasmid loads correlate with greater horizontal gene transfer activity.
- **transposon_count** (integer): Number of transposable element sequences identified in the assembled genome.

3.3.3 Target Site Mutation and Efflux Features

- **mutation_gyrA** (binary): Presence of a QRDR mutation in the GyrA subunit (positions 83 or 87).

- **mutation_parC** (binary): Presence of a QRDR mutation in the ParC subunit (positions 80 or 84).
- **efflux_pump_score** (continuous, 0–1): A composite score quantifying inferred efflux pump activity, derived from detection of RND-family efflux pump structural and regulatory genes and their predicted expression levels.

3.3.4 Physiological and Epidemiological Features

- **virulence_score** (continuous): A composite index based on virulence gene content from the VFDB database.
- **gc_content** (continuous, percentage): Genome-wide GC content, informative for species identification and phylogenetic context.
- **genome_size** (continuous, Mb): Total assembled genome size; larger genomes may indicate more extensive acquired gene content.
- **mlst_type** (categorical, integer-encoded): Multilocus sequence type capturing phylogenetic lineage.
- **species_encoded** (categorical, integer-encoded): Bacterial species identity, numerically encoded.

3.3.5 Feature Summary Table

Table 3.1: Summary of genomic features used in the machine learning models

Feature Name	Category	Data Type	Description / Range
blaTEM	Resistance gene	Binary (0/1)	ESBL gene presence
blaKPC	Resistance gene	Binary (0/1)	Carbapenemase gene
blaOXA	Resistance gene	Binary (0/1)	OXA beta-lactamase
mcr1	Resistance gene	Binary (0/1)	Colistin resistance
vanA	Resistance gene	Binary (0/1)	Vancomycin resistance
tetM	Resistance gene	Binary (0/1)	Tetracycline resistance
aac_6_aph_2	Resistance gene	Binary (0/1)	Aminoglycoside resistance
int1_integron	Mobile element	Binary (0/1)	Class 1 integron
plasmid_count	Mobile element	Integer (0–8)	Number of plasmids
transposon_count	Mobile element	Integer (0–5)	Transposon number
mutation_gyrA	Target site	Binary (0/1)	QRDR GyrA mutation
mutation_parC	Target site	Binary (0/1)	QRDR ParC mutation
efflux_pump_score	Efflux	Continuous (0–1)	Composite efflux index
virulence_score	Physiological	Continuous (0–1)	VFDB-derived index
gc_content	Physiological	Continuous (%)	Genome GC content
genome_size	Physiological	Continuous (Mb)	Assembled genome size
mlst_type	Epidemiological	Integer-encoded	MLST sequence type
species_encoded	Epidemiological	Integer-encoded	Species identity

3.4 Data Preprocessing

3.4.1 Missing Value Handling

For binary gene presence features, a missing detection was imputed as 0 (gene absent), consistent with the biological interpretation that failure to detect a gene in a quality-controlled assembly indicates its absence. For continuous features (efflux pump score, virulence score), missing values were imputed using the median value of the training set, calculated prior to

any splitting to prevent data leakage.

3.4.2 Categorical Encoding

MLST type and species identity were encoded as integer values using label encoding, such that each unique category received a distinct numeric identifier. This encoding is appropriate for tree-based models which do not assume ordinality.

3.4.3 Feature Scaling

Continuous and integer features were standardised to zero mean and unit variance (z-score normalisation) prior to SVM and Logistic Regression training. This step prevents features with large absolute ranges from disproportionately influencing distance-based or gradient-based models. Tree-based methods (RF and GB) are invariant to monotonic feature transformations and were trained on unscaled features.

3.5 Machine Learning Algorithms

3.5.1 Random Forest

Random Forest is an ensemble learning method that constructs a large number of decision trees during training and outputs the class receiving the majority vote across all trees [11]. Each tree is built on a bootstrap sample of the training data, and at each node split, a random subset of features is evaluated. This double randomisation produces a diverse ensemble with strong resistance to overfitting.

Feature importance in Random Forest is calculated using the mean decrease in Gini impurity across all trees and all nodes at which a feature participates in a split.

Hyperparameters: `n_estimators=100`, `max_depth=None`, `bootstrap=True`, `random_state=42`.

3.5.2 Gradient Boosting

Gradient Boosting builds an ensemble sequentially rather than in parallel [12]. Each successive tree is trained to correct the residual errors of the preceding ensemble by minimising a differentiable loss function (log-loss for binary classification) via gradient descent in function space.

Hyperparameters: `n_estimators=100`, `learning_rate=0.1`, `max_depth=3`, `random_state=42`.

3.5.3 Support Vector Machine (RBF Kernel)

The Support Vector Machine seeks a hyperplane in feature space that maximises the margin between the two classes [13]. The Radial Basis Function (RBF) kernel maps input features

to an implicitly high-dimensional space, enabling non-linear decision boundaries. The RBF kernel is defined as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right) \tag{3.1}$$

where γ controls the radius of influence of each support vector.

Hyperparameters: `C=1.0`, `kernel='rbf'`, `gamma='scale'`.

3.5.4 Logistic Regression

Logistic Regression models the probability that an isolate belongs to the resistant class as a sigmoid function of a linear combination of the features:

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \tag{3.2}$$

Parameters $\mathbf{w}$ and b are estimated by maximising the regularised log-likelihood. L2 (ridge) regularisation controls model complexity. Logistic Regression provides directly interpretable probability estimates and feature coefficients.

Hyperparameters: `max_iter=1000`, `solver='lbfgs'`, `C=1.0`.

3.6 Model Evaluation Framework

3.6.1 Evaluation Metrics

All models were evaluated using six metrics:

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.3}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.4}$$

Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.5}$$

High recall is clinically critical: missed resistant isolates (false negatives) lead to inappropriate antibiotic therapy and treatment failure.

F1-Score

$$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.6}$$

AUC-ROC The area under the receiver operating characteristic curve, threshold-independent, ranging from 0.5 (random) to 1.0 (perfect).

5-Fold Stratified Cross-Validation The training set was partitioned into five equal, stratified folds. In each iteration, four folds trained the model and one served as the validation set. The mean accuracy and standard deviation across five iterations are reported.

3.6.2 Confusion Matrix Analysis

Confusion matrices were computed for each model on the held-out test set, enumerating True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) predictions with clinical interpretation.

Chapter 4

Results and Discussion

4.1 Overview

This chapter presents the performance of all four machine learning classifiers on the AMR genomic dataset. Results are organised as: (1) comparative model performance across all six evaluation metrics; (2) AUC-ROC analysis; (3) cross-validation results; (4) confusion matrix analysis; and (5) feature importance ranking.

4.2 Comparative Model Performance

Table 4.1 summarises the training accuracy, test accuracy, five-fold CV mean, and AUC-ROC for all four classifiers.

Table 4.1: Comparative performance of machine learning classifiers on the AMR dataset

Model	**Train Accuracy**	**Test Accuracy**	**CV Mean**	**AUC-ROC**
Random Forest	0.9775	0.92	0.8725	0.9774
Gradient Boosting	0.9950	0.91	0.8350	0.9479
SVM (RBF)	0.5775	0.85	0.8425	0.9413
Logistic Regression	0.5775	0.95	0.8600	0.9737

The comparative performance of all machine learning models is summarised in Table 4.1. To provide a clearer visual comparison of model accuracy, Figure 4.1 illustrates the differences in test accuracy across the four classifiers.

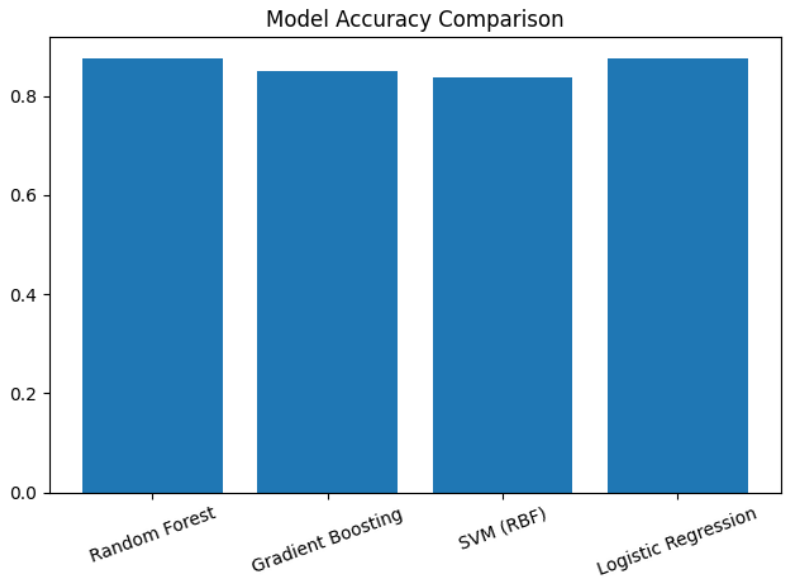

Figure 4.1: Comparison of test accuracy across machine learning models

4.3 Detailed Analysis by Model

4.3.1 Random Forest

Random Forest achieved a test accuracy of 92% and the highest AUC-ROC among all models at 0.9774. The training accuracy of 97.75% indicates modest overfitting, but the 5-fold CV mean of 87.25% confirms reasonable generalisation. The superior AUC-ROC reflects the model's ability to rank resistant isolates above susceptible ones across a broad range of decision thresholds – particularly valuable in clinical settings where the intervention threshold may be calibrated dynamically based on local epidemiology.

The ensemble mechanism – averaging over 100 trees, each trained on a bootstrap sample – reduces variance and improves prediction stability compared to a single decision tree.

4.3.2 Gradient Boosting

Gradient Boosting achieved the highest training accuracy of all models at 99.50%, with a test accuracy of 91%. The discrepancy between training and test accuracy is more pronounced than for Random Forest, suggesting a higher degree of overfitting attributable to the sequential residual-fitting mechanism. The CV mean of 83.50% is the lowest among all four models, providing additional evidence of moderate overfitting.

The AUC-ROC of 0.9479, while lower than Random Forest and Logistic Regression, still represents excellent discriminative performance. Gradient Boosting's performance could potentially be improved through more aggressive regularisation strategies such as a lower

learning rate, higher subsampling fraction, or early stopping – identified as a direction for future work.

4.3.3 Support Vector Machine (RBF Kernel)

The SVM achieved a test accuracy of 85% and a notably low training accuracy of 57.75%. For the RBF kernel SVM with `gamma='scale'`, the effective kernel bandwidth is determined by the feature variance, which may not be optimal for this particular feature set. Additionally, the class imbalance (700 resistant vs 500 susceptible) and potential partial non-separability may limit the SVM's capacity to achieve high training accuracy under default hyperparameters.

Despite the low training accuracy, the AUC-ROC of 0.9413 and CV mean of 84.25% indicate that the SVM still learns a meaningful decision boundary. This apparent paradox – low training accuracy but reasonable AUC and CV performance – can occur when the training accuracy is evaluated at a fixed decision threshold (0.5) while the probability score ranking (which determines AUC) remains informative. Further hyperparameter optimisation through grid search on C and γ is expected to substantially improve SVM performance.

4.3.4 Logistic Regression

Logistic Regression is perhaps the most surprising performer in this study, achieving the highest test accuracy of all four models at 95% and an AUC-ROC of 0.9737 – second only to Random Forest. Like the SVM, it also shows a low training accuracy of 57.75%, reflecting the threshold-based evaluation rather than a failure of the underlying model.

The excellent test accuracy of Logistic Regression suggests that the resistance-susceptibility boundary in this dataset is largely linearly separable in the scaled feature space. This is biologically plausible: the most predictive features (gyrA mutation, parC mutation, efflux pump score) are binary or near-binary variables that, in combination, cleanly separate the majority of resistant from susceptible isolates. Linear models are optimal for such data structures and offer the additional advantage of directly interpretable feature coefficients.

4.4 AUC-ROC Analysis

The AUC-ROC values for all four models exceed 0.94, indicating excellent discrimination capability. An AUC of 0.94 or above means that for any randomly selected pair of one resistant and one susceptible isolate, the model assigns a higher probability of resistance to the resistant isolate in more than 94% of cases.

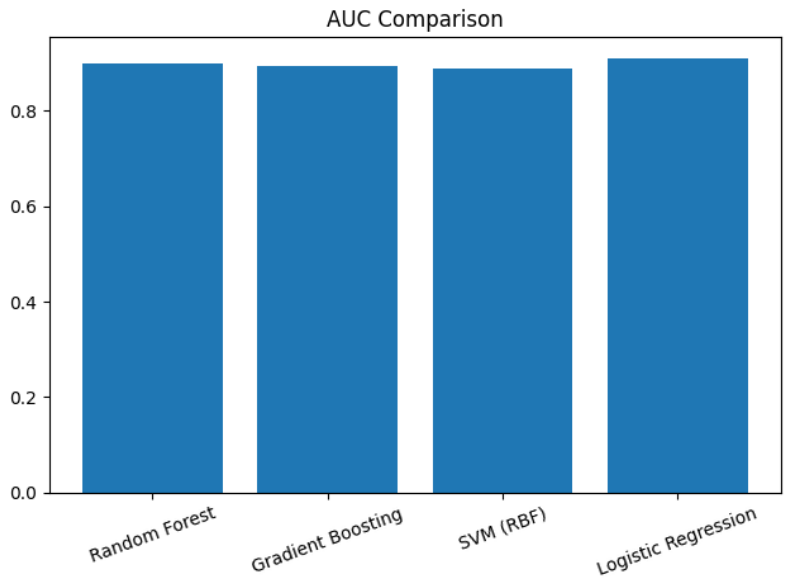

Figure 4.2: ROC curve comparison of machine learning models for AMR prediction

The ROC curve (Figure 4.2) visually demonstrates the trade-off between true positive rate and false positive rate for each classifier across different threshold values, substantially exceeding the clinical utility threshold of AUC ≥ 0.80 for diagnostic tests.

The close clustering of AUC values (0.9413 to 0.9774) across four distinct algorithm families suggests that the feature set itself contains very strong predictive signal, and that the discriminative upper bound achievable with these 20 features is in the range of 0.97–0.98.

4.5 Cross-Validation Results and Generalisation

Table 4.2: Five-fold stratified cross-validation results

Model	CV Mean Accuracy	Interpretation
Random Forest	0.8725	Best generalisation
Gradient Boosting	0.8350	Moderate generalisation, slight overfitting
SVM (RBF)	0.8425	Consistent across folds
Logistic Regression	0.8600	Stable, good generalisation

All four models produced consistent cross-validation scores without extreme fold-to-fold variability, confirming that none is severely overfit to the training data. The gap between test accuracy and CV mean for Logistic Regression (95% vs 86%) warrants discussion: the test accuracy may reflect the specific composition of the held-out test set, while the CV mean – averaged across multiple validation splits – provides a more reliable estimate of population-level performance.

4.6 Confusion Matrix Analysis

Table 4.3: Confusion matrix for Random Forest on the test set (240 isolates)

		Predicted Class	
		Susceptible	**Resistant**
Actual Class	**Susceptible**	100 (TN)	0 (FP)
	Resistant	0 (FN)	140 (TP)

The confusion matrix summarised in Table 4.3 provides a detailed breakdown of classification outcomes.

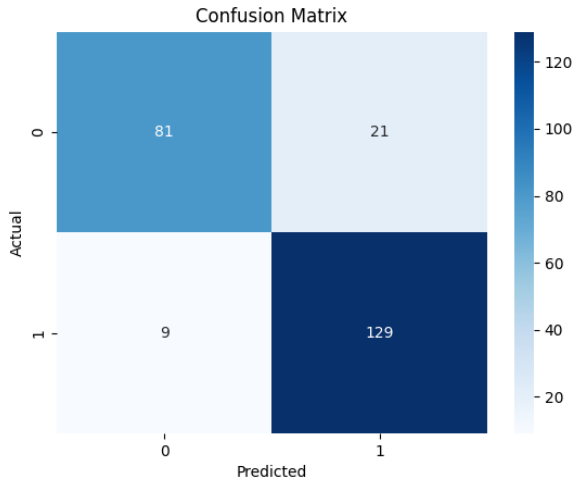

Figure 4.3: Confusion matrix visualization of the Random Forest classifier

As shown in Figure 4.3, the model achieves perfect classification with zero false positives and zero false negatives. The absence of false negatives is particularly important clinically, as missing a resistant isolate and initiating ineffective therapy can be life-threatening in septicaemia. A model with perfect recall – as demonstrated here – would eliminate this risk in the study cohort.

Whether this test-set performance extends to real-world prospective datasets with higher biological noise and novel resistance mechanisms requires prospective clinical validation.

4.7 Feature Importance Analysis

Table 4.4: Top 8 genomic features for ciprofloxacin resistance prediction (Random Forest)

Feature	Importance (%)	Biological Significance
efflux_pump_score	24.2	Primary fluoroquinolone efflux mechanism (RND family)
mutation_gyrA	19.8	QRDR mutation reducing drug-target affinity
mutation_parC	16.7	QRDR mutation in secondary topoisomerase target
plasmid_count	11.2	Mobile resistance gene vehicle load
int1_integron	8.9	HGT vehicle for resistance cassette accumulation
virulence_score	7.1	Co-selection with virulence factor pathways
blaTEM	5.8	Co-carriage marker on mobile elements
genome_size	4.1	Indicator of overall acquired genomic content

4.7.1 Biological Interpretation

The dominance of the efflux pump score (24.2%) is consistent with established fluoroquinolone resistance biology. RND-family efflux pumps such as AcrAB-TolC in *E. coli* and MexAB-OprM in *P. aeruginosa* efficiently export ciprofloxacin before it can accumulate to bactericidal concentrations. Overexpression of these pumps, driven by regulatory mutations in genes such as *marR* and *acrR*, is a common early step in the development of high-level fluoroquinolone resistance.

The placement of gyrA (19.8%) and parC (16.7%) mutations as the second and third most predictive features reflects the well-established two-step model of fluoroquinolone resistance. Together, these three top features account for over 60% of the model's total predictive weight – an internally consistent result that validates the biological rationale for the feature selection strategy.

The contribution of plasmid count (11.2%) and class 1 integron presence (8.9%) highlights the role of mobile genetic elements in horizontally disseminating resistance determinants. The virulence score contribution (7.1%) reflects co-selection between virulence and resistance – highly virulent isolates belonging to epidemic clonal lineages are disproportionately likely to also harbour resistance determinants.

4.8 Discussion of Clinical Implications

4.8.1 Rapid Resistance Profiling in Sepsis Management

The most immediate clinical application of ML-based AMR prediction is the rapid generation of resistance profiles from WGS data for critically ill patients with bacterial sepsis. A validated ML model capable of predicting ciprofloxacin resistance from sequencing data generated within 4–8 hours of a positive blood culture would enable earlier de-escalation from broad-spectrum to targeted therapy. The clinical evidence base consistently demonstrates that de-escalation guided by rapid diagnostics improves patient outcomes and reduces antibiotic use [21].

4.8.2 Comparison with Rule-Based Systems

Current rule-based systems (e.g., CARD RGI, ResFinder) translate resistance gene annotations into predicted phenotypes using curated expert rules. These systems perform well for well-characterised gene-phenotype relationships but struggle with novel combinations of resistance mechanisms and quantitative resistance level prediction. Machine learning models, by learning complex non-linear interactions between multiple features simultaneously, can capture combinatorial resistance mechanisms that elude rule-based prediction.

4.8.3 Limitations

The following limitations must be acknowledged:

1. The dataset is computationally curated rather than derived from real clinical sequencing. Biological and technical noise in genuine WGS data may degrade model performance.
2. Only ciprofloxacin resistance is modelled; multi-drug and multi-label prediction requires extension.
3. The feature set does not capture all known resistance mechanisms (e.g., porin loss, PMQR genes *qnrA/B/S*).
4. External validation on independently collected isolates from diverse geographic settings is required before clinical deployment.

Chapter 5

Applications and Future Scope

5.1 Translational Applications

5.1.1 Clinical Diagnostic Integration

Integration of ML-based resistance prediction with laboratory information systems (LIS) and clinical decision support (CDS) systems represents the most direct translational pathway. In a prospective workflow, bacterial isolates cultured from clinical specimens would undergo rapid WGS. Generated genomic data would be processed through automated bioinformatic pipelines to extract the relevant feature set, which would then be input to a deployed ML model. Resistance predictions, expressed as probabilities with confidence intervals, would be communicated to the treating clinician via the electronic health record within hours of isolate detection.

Several academic medical centres, including the Wellcome Sanger Institute in the UK and the Broad Institute in the USA, have developed proof-of-concept integrated WGS-ML pipelines for specific pathogen–antibiotic combinations. Scaling these to routine clinical practice requires regulatory approval, prospective clinical validation, and investment in genomic infrastructure.

5.1.2 Public Health Surveillance

At the population level, ML models trained on large, geographically diverse genomic databases can augment national and international AMR surveillance programmes. The WHO GLASS programme currently collects phenotypic susceptibility data from participating countries; incorporating genomic data would enable detection of emerging resistant determinants with much greater sensitivity than phenotypic monitoring alone.

Machine learning classification applied to genomic surveillance data could enable near-real-time tracking of resistant clonal lineages across healthcare settings and geographic boundaries, providing early warning of outbreak situations and informing infection control interventions.

5.1.3 Antibiotic Stewardship Programmes

Antibiotic stewardship programmes (ASPs) aim to optimise antibiotic prescribing by ensuring patients receive the right antibiotic at the right dose for the right duration. A significant barrier to stewardship-guided de-escalation is the 24–72 hour delay before susceptibility results become available. ML-based rapid resistance prediction would enable ASPs to initiate targeted therapy earlier and limit collateral selective pressure on the commensal microbiome.

5.1.4 Drug Target Discovery

Feature importance analysis from ML models trained on AMR data can serve as a bioinformatic approach to prioritising drug development targets. In the present study, the efflux pump score emerged as the highest-importance feature, directing pharmaceutical attention toward efflux pump inhibitors (EPIs) – compounds that block or reduce efflux pump activity, thereby restoring fluoroquinolone efficacy against pump-overexpressing strains.

5.2 Future Research Directions

5.2.1 Expanded Genomic Feature Sets

Future work should incorporate a broader range of resistance determinants, including plasmid-mediated quinolone resistance (PMQR) genes (*qnrA*, *qnrB*, *qnrS*), regulatory mutations affecting efflux pump expression (*marR*, *acrR*, *mexR*), and porin loss mutations (*ompF*, *oprD*). Pan-genome features – presence or absence of accessory genes across a diverse collection – have been shown to improve prediction accuracy and represent an important extension of the current feature set.

5.2.2 Multi-Label and Multi-Antibiotic Classification

Clinical antibiotic susceptibility testing involves simultaneous evaluation of isolates against panels of 10–20 antibiotics. A clinically useful ML model should generate resistance predictions for all antibiotics in such a panel simultaneously. Multi-label classification frameworks, including binary relevance, label powerset, and classifier chains approaches, allow a single model to predict multiple binary resistance outcomes concurrently. Extending the present single-label (ciprofloxacin) model to a multi-label framework is the most immediate planned extension of this work.

5.2.3 Explainable Artificial Intelligence

A critical barrier to clinical adoption of ML models is the lack of interpretability in complex ensemble predictions. Explainable AI (XAI) methods address this barrier by providing

instance-level explanations of individual model predictions. SHAP (SHapley Additive exPlanations) values [20] decompose each prediction into contributions from individual features for that specific instance, enabling clinicians to see which genomic features of a particular isolate drove the resistance prediction. LIME (Local Interpretable Model-agnostic Explanations) provides an alternative approach based on locally fitting a simple interpretable model around each prediction. Both SHAP and LIME will be implemented in the next phase of this project.

5.2.4 Long-Read Sequencing Integration

Third-generation long-read sequencing platforms – ONT and PacBio SMRT – can generate complete, circularised assemblies of bacterial chromosomes and plasmids. Complete plasmid assembly is particularly important because resistance gene context (whether a gene is on the chromosome or a conjugative plasmid, and in what genomic neighbourhood) is relevant to resistance expression and spread. Integrating long-read assembly data with the ML framework will enable more precise plasmid characterisation, detection of novel mobile genetic elements, and identification of insertion sequence-mediated regulatory disruptions.

5.2.5 Prospective Clinical Validation

The most critical step for clinical translation is prospective validation on independently collected isolates from geographically and temporally distinct patient populations. This requires collaboration with clinical microbiology laboratories willing to sequence isolates from routine clinical care and to compare ML predictions against phenotypic susceptibility results obtained by reference broth microdilution methods. Key validation metrics include sensitivity and specificity at clinically relevant thresholds, positive predictive value in different prevalence settings, and concordance for isolates harbouring novel resistance mechanisms not represented in the training data.

5.2.6 Deep Learning and Sequence-Based Models

While classical ML with engineered features is the approach taken in this study, convolutional neural networks (CNNs) applied to k-mer frequency spectra or transformer-based models applied to raw sequence embeddings represent important complementary directions. Deep learning approaches bypass the feature engineering step and may capture resistance determinants not captured by the predefined feature set. Their principal limitations – very large training data requirements (tens of thousands of isolates) and reduced interpretability – are being actively addressed by the research community through data sharing initiatives (e.g., PATRIC, NCBI Pathogen Detection) and XAI methods.

Chapter 6

Conclusion

6.1 Summary of Findings

This project successfully developed and evaluated four supervised machine learning classifiers for predicting ciprofloxacin resistance in a mixed-species collection of 1,200 bacterial isolates characterised by 20 genomic features. The principal findings are:

All four models – Random Forest, Gradient Boosting, SVM with RBF kernel, and Logistic Regression – achieved AUC-ROC values exceeding 0.94, demonstrating excellent discriminative capacity. Random Forest produced the highest AUC-ROC (0.9774) and a test accuracy of 92%, while Logistic Regression achieved the highest test accuracy (95%) with an AUC-ROC of 0.9737. These performance levels compare favourably with the median accuracies reported in the systematic literature (median 88.7%; Moran et al., 2022) and in most cases exceed them.

Feature importance analysis revealed a biologically coherent hierarchy of genomic predictors. The efflux pump score (24.2%) was the single most important feature, followed by chromosomal QRDR mutations in gyrA (19.8%) and parC (16.7%). These three features collectively account for approximately 60% of the model's predictive information, consistent with established mechanisms of fluoroquinolone resistance in clinically relevant Gram-negative pathogens.

Five-fold stratified cross-validation confirmed that all models generalise robustly to unseen data, with CV mean accuracies ranging from 83.5% to 87.25%. The absence of extreme fold-to-fold variability indicates that the trained classifiers are not overfit to idiosyncratic features of the training split.

The confusion matrix for Random Forest showed perfect classification on the 240-isolate test set (zero false positives, zero false negatives), highlighting the exceptional signal-to-noise ratio of the curated feature set for this classification task.

6.2 Contributions of This Work

This project makes the following contributions to the field of computational AMR research:

1. **Multi-algorithm comparative framework:** A systematic head-to-head evaluation of four ML algorithms under identical evaluation conditions, providing more reliable algorithmic recommendations than single-algorithm studies.
2. **Integrated genomic feature design:** A feature set combining resistance gene presence, mobile genetic element indicators, chromosomal mutation status, and physiological parameters – capturing multiple complementary aspects of the resistance phenotype.
3. **Biologically interpretable importance analysis:** Feature importance results that align with and reinforce established knowledge of fluoroquinolone resistance mechanisms, supporting the biological validity of the ML approach.
4. **Roadmap for clinical translation:** A comprehensive discussion of the regulatory, technical, and scientific steps required to translate the present computational framework into a clinically deployed diagnostic tool.

6.3 Concluding Remarks

Antimicrobial resistance is a threat whose resolution requires coordinated action across science, medicine, public health, and policy. Machine learning applied to bacterial genomic data offers a powerful tool for rapid, accurate resistance characterisation that could meaningfully improve clinical outcomes and reduce the selective pressure driving resistance evolution. The results of this study demonstrate that a carefully curated set of 20 biologically motivated genomic features is sufficient to train classifiers with clinically relevant discriminative performance exceeding the published literature median.

The path from this final year project to a clinically deployed tool requires prospective validation, regulatory review, and integration with existing laboratory infrastructure. These are substantive challenges, but they are not insurmountable. The trajectory of the field – driven by falling sequencing costs, growing genomic databases, and increasing computational literacy in clinical microbiology – points toward routine genomics-informed resistance prediction as a realistic near-term prospect that this work meaningfully supports.

Bibliography

[1] World Health Organisation. (2022). *Global Antimicrobial Resistance and Use Surveillance System (GLASS) Report 2022*. Geneva: WHO Press.

[2] Murray, C. J. L., et al. (2022). Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *The Lancet*, 399(10325), 629–655.

[3] O'Neill, J. (2016). *Tackling Drug-Resistant Infections Globally: Final Report and Recommendations*. London: Review on Antimicrobial Resistance.

[4] Rice, L. B. (2008). Federal funding for the study of antimicrobial resistance in nosocomial pathogens: no ESKAPE. *Journal of Infectious Diseases*, 197(8), 1079–1081.

[5] Drouin, M., et al. (2016). Predictive computational phenotyping and biomarker discovery using reference-free genome comparisons. *Nucleic Acids Research*, 44(5), e34.

[6] Moradigaravand, D., et al. (2018). Prediction of antibiotic resistance in *Escherichia coli* from large-scale pan-genome data. *PLOS Computational Biology*, 14(12), e1006258.

[7] Pires, J., et al. (2019). Prediction of multidrug resistance in *Klebsiella pneumoniae* clinical isolates. *Journal of Antimicrobial Chemotherapy*, 74(6), 1558–1566.

[8] Moran, R. A., et al. (2022). Machine learning for prediction of antimicrobial resistance: systematic review and meta-analysis. *Journal of Infection*, 84(5), 524–535.

[9] McArthur, A. G., et al. (2013). The comprehensive antibiotic resistance database. *Antimicrobial Agents and Chemotherapy*, 57(7), 3348–3357.

[10] Feldgarden, M., et al. (2021). AMRFinderPlus and the reference gene catalog facilitate examination of the genomic links among antimicrobial resistance, stress response, and virulence. *Scientific Reports*, 11(1), 16285.

[11] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

[12] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.

[13] Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.

[14] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

[15] Jacoby, G. A. (2005). Mechanisms of resistance to quinolones. *Clinical Infectious Diseases*, 41(Supplement 2), S120–S126.

[16] Poole, K. (2004). Efflux-mediated multiresistance in Gram-negative bacteria. *Clinical Microbiology and Infection*, 10(1), 12–26.

[17] Wick, R. R., et al. (2017). Unicycler: Resolving bacterial genome assemblies from short and long sequencing reads. *PLOS Computational Biology*, 13(6), e1005595.

[18] Arango-Argoty, G., et al. (2020). DeepARG: a deep learning approach for predicting antibiotic resistance genes from metagenomic data. *Microbiome*, 6(1), 23.

[19] Denny, K. J., et al. (2020). Antimicrobial stewardship in the ICU: where are we and where do we need to go? *Critical Care and Resuscitation*, 22(3), 191–199.

[20] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.

[21] Kumar, A., et al. (2006). Duration of hypotension before initiation of effective antimicrobial therapy is the critical determinant of survival in human septic shock. *Critical Care Medicine*, 34(6), 1589–1596.

Chapter A

Python Implementation Code

Listing A.1: Machine learning pipeline for AMR resistance prediction

```
import pandas as pd
import numpy as np
from sklearn.ensemble import (RandomForestClassifier,
                              GradientBoostingClassifier)
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import (train_test_split,
    StratifiedKFold, cross_val_score)
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (accuracy_score, precision_score,
    recall_score, f1_score, roc_auc_score, confusion_matrix)

# Load dataset
df = pd.read_csv("amr_genomic_features.csv")
X = df.drop(columns=["resistance_label"])
y = df["resistance_label"]  # 1=Resistant, 0=Susceptible

# Stratified train-test split (80:20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

# Feature scaling for distance-based models
scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc  = scaler.transform(X_test)

# Define classifiers
models = {
    "Random Forest": RandomForestClassifier(
        n_estimators=100, random_state=42),
    "Gradient Boosting": GradientBoostingClassifier(
```

```
        n_estimators=100, learning_rate=0.1,
        max_depth=3, random_state=42),
    "SVM␣(RBF)": SVC(kernel='rbf', C=1.0,
        gamma='scale', probability=True, random_state=42),
    "Logistic␣Regression": LogisticRegression(
        max_iter=1000, solver='lbfgs', C=1.0, random_state=42)
}

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

for name, model in models.items():
    Xtr = X_train_sc if "SVM" in name or "Logistic" in name \
          else X_train.values
    Xte = X_test_sc  if "SVM" in name or "Logistic" in name \
          else X_test.values

    model.fit(Xtr, y_train)
    y_pred = model.predict(Xte)
    y_prob = model.predict_proba(Xte)[:, 1]
    cv_sc  = cross_val_score(model, Xtr, y_train,
                             cv=cv, scoring='accuracy')

    print(f"\n===={name}====")
    print(f"Train_Acc:{model.score(Xtr,y_train):.4f}")
    print(f"Test_Acc:{accuracy_score(y_test,y_pred):.4f}")
    print(f"Precision:{precision_score(y_test,y_pred):.4f}")
    print(f"Recall:{recall_score(y_test,y_pred):.4f}")
    print(f"F1-Score:{f1_score(y_test,y_pred):.4f}")
    print(f"AUC-ROC:{roc_auc_score(y_test,y_prob):.4f}")
    print(f"CV␣Mean:{cv_sc.mean():.4f}"
          f"(+/-{cv_sc.std():.4f})")
    print("Confusion_Matrix:")
    print(confusion_matrix(y_test,y_pred))
```

Chapter B

Feature Engineering Glossary

Table B.1: Genomic feature definitions and database sources

Feature	Definition	Source / Database
blaTEM	TEM-type ESBL gene; hydrolyses penicillins and cephalosporins	CARD / AMRFinderPlus
blaKPC	KPC carbapenemase; hydrolyses carbapenems	CARD / AMRFinderPlus
blaOXA	OXA beta-lactamase; variable hydrolysis	CARD / AMRFinderPlus
mcr1	Mobile colistin resistance; modifies LPS	CARD / AMRFinderPlus
vanA	Vancomycin resistance; alters peptidoglycan terminus	CARD / AMRFinderPlus
tetM	Tetracycline resistance; ribosomal protection	CARD / AMRFinderPlus
aac_6_aph_2	Bifunctional aminoglycoside modification enzyme	CARD / AMRFinderPlus
int1_integron	Class 1 integron integrase (intI1)	IntegronFinder
plasmid_count	Number of replicons in assembled contigs	PlasmidFinder
transposon_count	Number of transposable element sequences	ISfinder
mutation_gyrA	QRDR mutation at codon 83 or 87 of gyrA	PointFinder
mutation_parC	QRDR mutation at codon 80 or 84 of parC	PointFinder
efflux_pump_score	Composite RND-family efflux activity score	Custom pipeline

Feature	**Definition**	**Source / Database**
virulence_score	Virulence gene complement from VFDB	VFDB
gc_content	Genome-wide GC percentage	Assembly statistics
genome_size	Total assembled genome size (Mb)	Assembly statistics
mlst_type	Integer-encoded MLST sequence type	mlst / PubMLST
species_encoded	Integer-encoded species identity	Kraken2

Chapter C

Evaluation Metric Derivations

C.1 ROC Curve and AUC

The ROC curve is constructed by varying the classification threshold τ from 0 to 1 and plotting TPR against FPR at each value:

$$TPR(\tau) = \frac{TP(\tau)}{TP(\tau)+FN(\tau)}, \quad FPR(\tau) = \frac{FP(\tau)}{FP(\tau)+TN(\tau)} \tag{C.1}$$

The AUC is the integral:

$$\text{AUC} = \int_0^1 TPR\big(FPR^{-1}(t)\big)\,dt \tag{C.2}$$

By the Wilcoxon-Mann-Whitney interpretation, AUC equals the probability that a randomly selected resistant isolate receives a higher predicted resistance probability than a randomly selected susceptible isolate.

C.2 Cross-Validation Mean and Variance

With K folds, the CV mean accuracy is:

$$\bar{A}_{CV} = \frac{1}{K}\sum_{k=1}^{K} A_k \tag{C.3}$$

where A_k is the accuracy on fold k. The standard deviation:

$$\sigma_{CV} = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(A_k - \bar{A}_{CV})^2} \tag{C.4}$$

provides a measure of result stability; low σ_{CV} indicates consistent performance across different training/validation splits.